

Access Control Cabling

Reader Cable Types

Access control readers typically communicate over Wiegand or OSDP. Wiegand uses unshielded twisted-pair for data and a separate 18 AWG pair for 12 VDC power. OSDP operates over RS-485 with a recommended shielded 22 AWG pair.

- Wiegand data: UTP
- Wiegand power: 18 AWG
- OSDP: RS-485, shielded 22 AWG

Door Position Monitoring

Door contacts signal an open or closed state to the access control panel. The monitoring loop must be supervised with an end-of-line resistor, typically 1 kOhm, located at the door contact.

- EOL resistor: 1 kOhm
- Located at door contact

Lock Power Budget

A typical electrified strike draws 250 mA at 12 VDC. A magnetic lock draws 500 mA at 12 VDC. Power supplies must be sized at 125 % of the calculated continuous load.